FIRE FIGHTER

Fire Extinguishing Competition

Task:

Create an autonomous robot that can locate and extinguish 4 candles placed in the arena, without touching them, inside a field outlined by a black line in the least possible time.

Aim:

Increase awareness and implementation of various sensors and components that can be used in robots to solve real-world problems.

FIRE FIGHTER

General rules:

1. All teams should consist only of 4 members (including team leader).
2. All team members should have Physical/college ID cards, which are to be presented for verification.
3. Only one bot can be registered for only one team.
4. No external power supply will be provided. Batteries should be placed inside the robot.
5. No damage should be made to the arena either by team member or team's bot.
6. The final decision regarding any appeals is made by the coordinators. All complaints must be reported to the coordinators during the match or right after the ending of the match. Complaints filed later will not be accepted. The final decision regarding any dispute or inconsistencies, is always made by the Team RoboDarshan.

FIRE FIGHTER

Gameplay rules:

1. The Run

- Every team will have 3 runs. Each team can take sufficient time gap between their runs. Any runs requested after the event time frame will not be entertained.
- Maximum Time: 5 minutes
- Start: The robot will be placed in the designated “Home Zone”.
- Activation: Upon the coordinator’s signal, the participant activates the robot (single switch press).
- The robot must autonomously scan the arena to locate the heat/light signature.
- If the robot returns to the Home Zone after extinguishing all 4 candles, points will be awarded (refer to scoring rules).
- Completion: The run ends when all fires are extinguished or the maximum time is reached.

2. Extinguishing

- Process of extinguishing the lit candle: Entering the outline, extinguishing, and leaving the circle. During this time, robot cannot make contact with the candle.
- Robot is counted as inside of the circle when at least one wheel is in contact with the black line.
- If the robot’s extinguishing mechanism extinguishes other candles (other than the intended), then otherwise extinguished candle’s points will not be awarded. In such case, Time Bonus will not be awarded.

FIRE FIGHTER

3. Resets and Restarts

- Each team is allowed only 2 Restarts overall if the robot malfunctions immediately after the start inside the Home Zone.
- Touching the robot during a run will be considered as “Manual RESET”. The robot must be brought back to the Home Zone while the time goes on.
- Inside the Home Zone, touching the bot will be considered as Restart. Outside the Home Zone, it will be considered as “Manual RESET” and penalty will be imposed.
- Timeout: If the robot is stuck in one place for more than 30 seconds, then coordinator can call a “Manual RESET” (with penalty).

FIRE FIGHTER

Arena Specifications:

1. Layout

- Arena Size (L x W): 2 meters x 2 meters.
- Boundaries: Enclosed by walls (approx. 15-20cm height) painted white or black (contrasting with the floor).
- Floor: Flat and smooth wooden or vinyl surface (non-reflective is preferred to avoid sensor confusion).
- Home Zone (L x W): 40cm x 40cm

2. Fire sources

- Source: A standard white wax candle (approx. heigh 10cm)
- Placement: 4 sources will be placed randomly in the arena.
- Safety circle: A black outline (radius 15cm) will be drawn around the candle. The robot must have one wheel touching the safety circle to successfully extinguish the candle).

FIRE FIGHTER

Robot Specifications:

1. Control System, Dimensions, and Power Source
   - Once turned on, the robot must be autonomous i.e. self-controlled without any human intervention.
   - Maximum dimensions allowed (L x W x H): 30cm x 30cm x 30cm
   - Maximum robot weight allowed: 2kg (including batteries)
   - The robot must be powered electrically.
   - Maximum voltage allowed is 12V - 24V (Standard Li-ion or Li-Po batteries).
   - Combustion engines or pneumatic sources requiring external compressors are strictly prohibited.

2. Extinguishing Mechanism
   - Method: The robot must use wind/air pressure (fans, blowers, propellers) to extinguish the fire.
   - Prohibited: Water, chemicals, CO2 sprays, or solid projectiles are not allowed.
   - Contact: The robot must not touch the fire source (candle/lamp). If the robot knocks the candle over, the run is immediately terminated.

FIRE FIGHTER

Scoring rules:

- Refer the scoring matrix as given below.
- Time starts with coordinator's signal and ends with the 4th candle extinguished. The remaining time will be considered as Time Bonus.
- Time bonus will be awarded upon coordinator's discretion.
- Previously extinguished candles become obstacles in the playfield, and will be counted as Wall Collision if touched.
- Return to Home points will be awarded only when all four wheels of the robot are in the Home Zone.
- In the final scorecard, the best score of the 3 runs will be considered. If any two teams have equal points, then time of the best run will be considered.
- **Formula: Total score = (Base score) + (50 * number of candles extinguished) – (Penalties) + (Time Bonus)**

FIRE FIGHTER

Criteria	Points
Base Score (To all teams)	100
Candle flame extinguished	+50 per candle
Return to Home Zone	+50
Time Bonus	+1 for every second saved (Total time – Actual time)
Wall Collision	-5 per collision
Candle Touched	-20 (if touched, not knocked over)
Knocking over candle	Run Disqualified
Manual RESET	-10 per touch

Disqualification will be awarded if:

- The robot uses dangerous materials (explosives, inflammables, etc.)
- The robot damages the arena floor or walls significantly and/or intentionally.
- The participants display unsportsmanlike behavior.
- The robot is not autonomous (hidden remote controls)

In case of any discrepancy the Decision of Team ROBODARSHAN will be final and binding

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